

ASSIGNMENT BRIEF

HTU Course No: 10203380

HTU Course Name: Computer Organization and Design

BTEC UNIT No: N/A

BTEC UNIT Name: N/A



Assignment Brief

Student Name/ID Number/Section	
HTU Course Number and Title	10203380 - Computer Organization and Design
BTEC Unit Number and Title	N/A
Academic Year	2025-2026 Fall
Assignment Author	Dr. Huthaifa Al-Omari
Course Tutor	Dr. Huthaifa Al-Omari, Eng. Asma Lafi
Assignment Title	Analysis and Design of Customized Computer System
Assignment Ref No.	1
Issue Date	Part 1: 11/12/2025 Part 2: 17/01/2026
Formative Assessment dates	12/12/2025 -to- 15/01/2026
Submission Date	Part 2: 17/01/2026 Part 2: 17/01/2026
IV Name & Date	Eng. Ashar Alkhamaiseh 30/11/2025

Submission Format

This assignment consists of two parts where the submission of the assignment should be:

For part 1:

An individual word-processed report that consists of:

- System Design
- Assembly language Coding Project using Mano's Simulator
- Problem Solving
- Research Task

Your report should be:

- written in a concise, formal business style using single spacing and font size 12 with use of headings, paragraphs and subsections as appropriate.
- supported with research and referenced using the Harvard referencing system.

Note: Handwritten text or hand-drawn figures are not allowed. All content must be typed, and all diagrams must be created using digital tools.

For part 2:

In-class closed book, closed notes examination.

Your answer should be clear and coherent. Providing final answers without showing detailed steps is not accepted. If the tasks are completed over multiple pages, ensure that your name and student number are present on each sheet of paper.

Unit Learning Outcomes

- LO1** Investigate the functions of computer system components.
- LO2** Examine the design and organization of basic computer.
- LO3** Discuss how data and programs can be represented within computer systems.
- LO4** Investigate advanced computer architecture and performance.

Assignment Brief and Guidance

You have recently been appointed as a Junior Systems Architect at TechWave Solutions, a company that designs custom computing systems for clients in sectors such as finance, healthcare, and engineering. Your manager has requested you prepare a technical report and system design proposal for a new customized computer. The client requires a custom-designed computing solution and wants clear justification of architectural choices, performance considerations, and the internal organization of the system components. This includes identifying the core components of the computer, explaining how instructions are processed, and presenting your findings using diagrams, examples, or a basic simulation. As a junior member of the development team, you are asked to solve the following problems to assess your proficiency in designing, coding, and validating the functionality and performance of the computer.

Part1:

- 1) Identify the main components of the Basic Computer (BC) that we built in the class and explain the function and the size of each component. Draw a diagram showing how the components are connected and organized.
- 2) Simplify the following Boolean functions using K-map and draw the logic diagrams for the obtained simplified expressions.
 - a. $f(W,X,Y,Z) = \sum m(0,2,3,4,5,7,8,10,13,15)$
 - b. $f(W,X,Y,Z) = \sum m(2,4,6,8,10,12)$
- 3) A computer uses a memory unit with 128K words of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts: an indirect bit, an operation code, a register code part to specify one of 33 registers, and an address part.
 - a. How many bits are there in the operation code, the register code part, and the address part?
 - b. Draw the instruction word format and indicate the number of bits in each part.
 - c. Indicate the size of each of the following registers by filling the following table.

Register	Size
IR	
TR	
AR	
AC	
DR	
PC	

- 4) Consider a computer system with the following characteristics: a memory unit that contains 32K words, with each word consisting of 16 bits, instruction code, where a single binary instruction is stored in one word of memory. The instruction format consists of: An indirect addressing mode bit, an operation code (opcode), with an unknown number of bits, A register selection code (4 bits) and an address part to access memory.
 - a. Determine the number of bits allocated for the operation code.
 - b. Draw the instruction word format, specifying the bit distribution for each component.
 - c. Design and illustrate the hardware decoding inputs from the instruction register to the control unit for this system.

- 5) The Basic Computer, as we discussed in the class, is a simple model that supports interrupt handling through a single interrupt. This system allows external devices to interrupt the normal execution of instructions, enabling input/output operations to be serviced efficiently. When an interrupt occurs, the Basic Computer temporarily halts the current program, saves the return address, and branches to an interrupt routine. Explain how the interrupt is handled by the Basic Computer. Your answer should be supported with flowcharts and a numerical example to demonstrate the steps involved in the interrupt cycle.

- 6) The ISZ instruction in the Basic Computer is used to Increment the contents of a memory location and Skip the next instruction if the result become Zero. List all the micro-operations involved in the execution of the ISZ instruction, covering the fetch, decode, and execute phases. Be sure to specify which registers and components are involved at each step, showing that in a detailed flowchart starting from time T_0 .

- 7) Design a common bus system for a digital system that contains seven general-purpose registers: R1, R2, R3, R4, R5, R6, and R7. Each register is 6 bits wide and must be able to transfer data to and from a shared bus. The system allows only one register to place data on the bus at any time, and only one register can load data from the bus at a time.

- 8) In digital computer systems, hexadecimal numbers are often used for compact representation of binary values. To interface with systems that operate on binary or octal bases, it is often necessary to convert hexadecimal numbers into these other numeral systems. Given the hexadecimal number FAE739BD45, convert it into its equivalent octal number.

- 9) Perform the following arithmetic operations with binary numbers in signed-2's complement representation. Use eight bits to accommodate each number together with its sign. Indicate whether an overflow occurs or not. Verify your answers.
 - a. $(+97) + (+120)$
 - b. $(-97) + (-120)$
 - c. $(+97) + (-120)$
 - d. $(-97) + (+120)$

10) In the Basic Computer (BC), arithmetic operations such as multiplication is not directly supported by hardware and must be implemented through step-by-step algorithms in assembly. Therefore, a programmer must implement multiplication using the available instructions. Develop an assembly program for the Basic Computer (BC) to perform the multiplication of two unsigned 8-bit numbers entered by the user. The result of the multiplication should be stored in a specific memory location. Explain in detail:

- a. How your program works.
- b. Provide a flowchart that shows the step-by-step procedure for programming the multiplication operation.
- c. Use the BC simulator to execute and verify your program. Your report must include screenshots illustrating how the program was validated.

11) Consider the following arithmetic expression:

$$\frac{\left[\frac{A}{B}\right] * \left[\frac{C}{D} + \frac{E}{F} - (G - H)\right]}{W + \left(\frac{X}{Y}\right)}$$

- a. Convert the above expression from infix to Reverse Polish notation.
- b. Show the stack operations for evaluating the expression.

12) Convert the following arithmetic expression from Reverse Polish notation to infix notation:

- a. A B C D E F G / - + * - *
- b. T U V W X Y Z * / * - / +

13) A non-pipeline system takes 50 ns to process a task. The same task can be processed in a 6-segment pipeline with a clock cycle of 8 ns. For number of tasks $n = 80$, find the following:

- a. Time required by the non-pipeline system to complete the tasks.
- b. Time required by the pipeline system to complete the tasks.
- c. Speedup ratio of the pipeline for the tasks.

- 14)** You are given a stream of data for which you must perform a sequence of arithmetic operations using a pipelined architecture. For each input set, the computation is defined as:

$$\frac{X_i}{Y_i} - \frac{W_i}{Z_i}, \quad \text{for } i = 1, 2, \dots, 7$$

Each arithmetic suboperation (division, subtraction) is executed in a separate pipeline segment. Registers R_1 through R_7 receive new input values at every clock cycle.

- a. Using Register Transfer Language (RTL), describe the micro-operations performed within each pipeline segment for the given expression. Then, provide a diagram that illustrates how data flows into and through the pipeline registers.
- b. Construct a Gantt-style timeline diagram illustrating the content and progression of values through the pipeline registers.
- c. Compute the total execution time for:
 - Non-pipelined execution
 - Pipelined execution
- d. Determine the pipeline speedup achieved.

- 15)** Conduct a detailed research study on Reduced Instruction Set Computer (RISC) and Complex Instruction Set Computer (CISC) architectures. Your report should explain the fundamental characteristics of each architecture and critically discuss their advantages and disadvantages. All information must be supported with appropriate sources, cited and referenced using the Harvard referencing style.

Part2:

An in-class exam on Saturday 17/01/2026

Learning Outcomes and Assessment Criteria

Pass		Merit	Distinction
LO1 Investigate the functions of computer system components			D1 Analyze digital components used in the organization of digital computers
P1 Identify the main subsystems of a computer and explain how they are organized and connected. P2 Carry out Boolean logic operations to simplify algebraic expressions in Boolean functions.	M1 Examine how K-Map is used to optimize the design of digital logic systems.		
LO2 Examine the design and organization of basic computer			D2 Create a design of control logic circuits for Basic Computer components.
P3 Examine the structure of instruction set for a basic digital computer. P4 Identify the sequence of events in carrying out the Instruction cycle.	M2 Examine the structure of the common bus system		
LO3 Discuss how data and programs can be represented within computer systems.			D3 Examine the content of memory and registers of the Basic Computer while executing low-level programs
P5 Investigate how different types of data can be converted and stored in computer systems. P6 Examine different types of addressing modes	M3 Create a low-level program which includes decision making, branching, subroutine, and I/O operations.		
LO4 Investigate advanced computer architectures and performance			D4 Examine the general register organization.
P7 Use stack organization to evaluate arithmetic expressions.	M4 Illustrate how pipelining improves the performance of a computer system. M5 Compare and contrast the RISC and the CISC architectures and show their advantages and disadvantages.		